



Treatment Interruption and Mortality among Persons Living with HIV Initiating Antiretroviral Treatment in Jamaica

Sharlene Beckford Jarrett¹, Sally Stephens¹, Tiffany Butterfield², Alisha Robb-Allen², Janeil Williams¹, Kelly-Ann Gordon-Johnson³, Ramanjot Kaur¹, Willi McFarland¹, Nicola Skyers²

¹ University of California, San Francisco; ² Ministry of Health and Wellness, Jamaica; ³ Centers for Disease Control and Prevention, Global Health Center, Division of Global HIV & TB Caribbean Office



Abstract

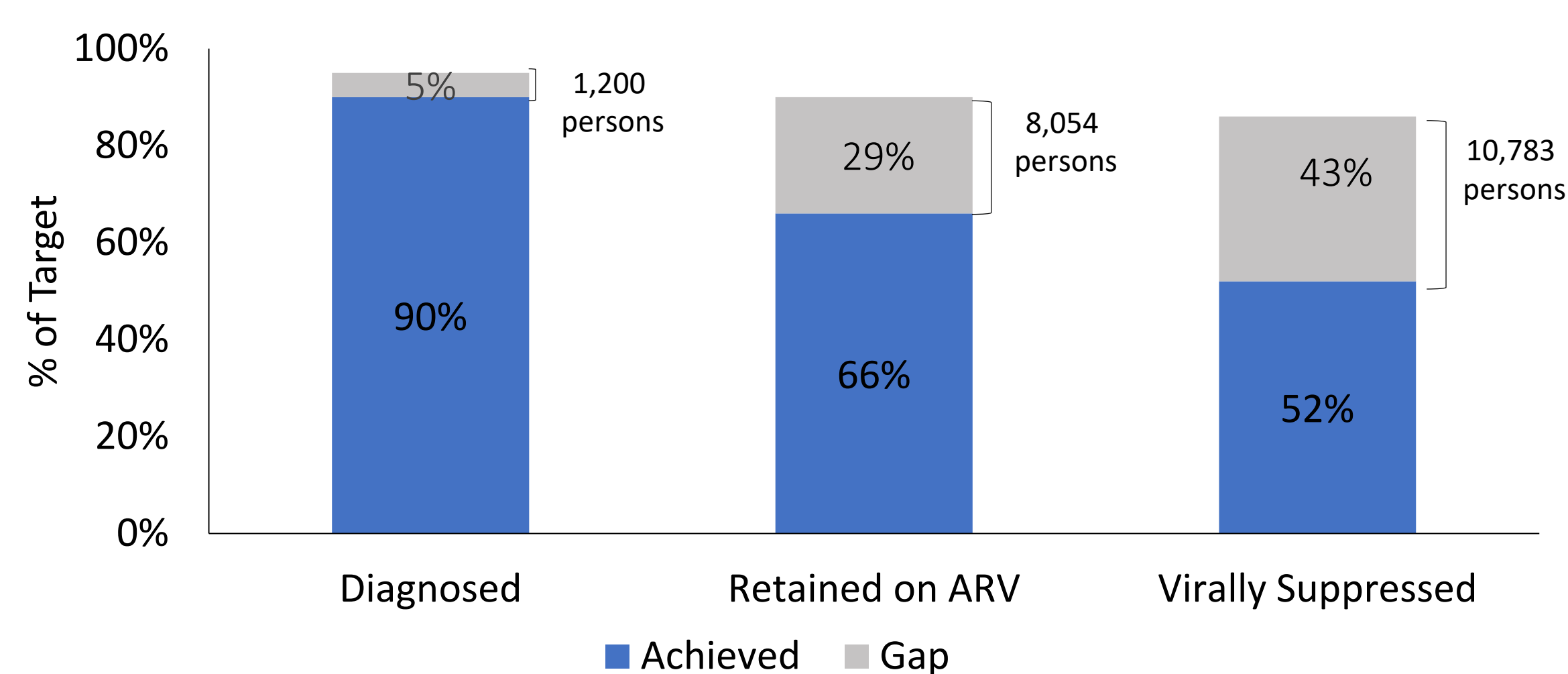
KEY FINDINGS:

- More than 25% of PLHIV are not retained on ART after 18 months after initiation
- Targeted support for clients aged ≥50 years, males, and persons initiating ART with low CD4 counts could avert mortality among persons with HIV

Introduction and Objectives

In 2020, 12,032 or 43% of persons living with HIV were retained on ART in Jamaica. In 2024, this number increased to 15,114 or 66% of persons living with HIV. Achieving epidemic control targets for Jamaica requires improved initiation and retention on lifesaving ART.

Figure 1. Achievement of the UNAIDS 95-95-95 Cascade among people living with HIV in Jamaica, 2024



Methods

- Retrospective cohort analysis of national data for 4,784 adults with HIV initiating ART January 2018 to December 2020 and followed up until June 2022.
- Kaplan-Meier, Cox proportional hazards, and logistic regression analyses identified factors associated with treatment interruption and mortality.
- Treatment interruption was defined as no clinic contact for ≥90 days after last scheduled appointment.

Results

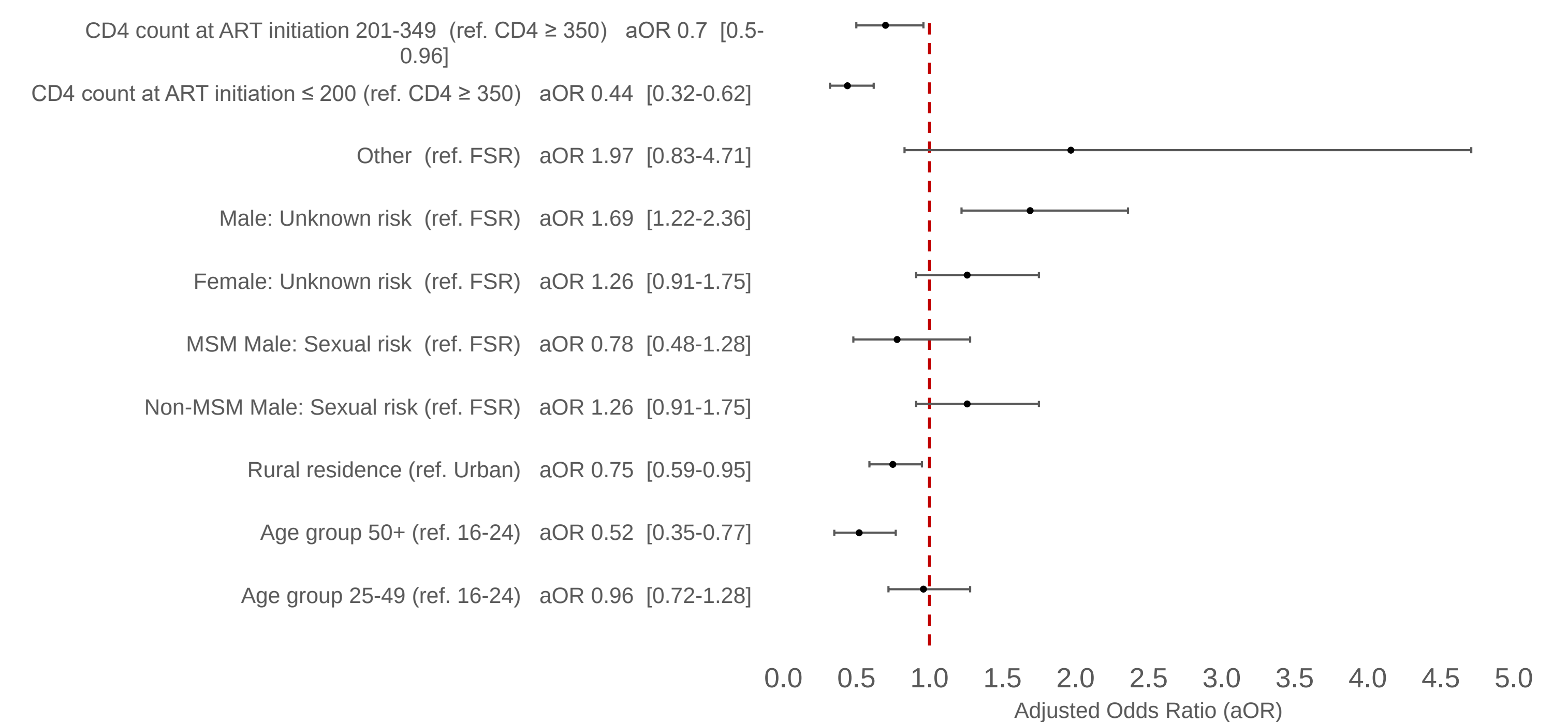
Table 1. Characteristics of 4784 patients initiating ART from January 1, 2018 to December 31, 2020, Jamaica

Characteristics	Total N (%)	Female n (%)	Male n (%)
Age (years) at ART initiation			
Median (IQR)	37 (27-48)	35 (26-47)	39 (28-49)
16 – 24	863 (18)	511 (20)	352 (15)
25 - 49	2,831 (59)	1,474 (59)	1,357 (60)
50+	1,090 (23)	523 (21)	567 (25)
CD4 count (cells/mL) at registration			
Median (IQR)	438 (232-643)	495 (279-706)	375(190-561)
≤200	693 (14.5)	308 (12.3)	385 (16.9)
201 - 349	507 (10.6)	231 (9.2)	276 (12.1)
≥350	1,910 (39.9)	1131 (45.1)	779 (34.2)
Unknown	1,674 (35.0)	838 (33.4)	836 (36.7)
Outcome			
Alive and on ART	3,554 (74)	1,930 (77)	1624 (71)
Treatment Interruption	540 (11)	273 (11)	267 (18)
Died	453 (10)	207 (8)	246 (11)
Migrated/transferred	237 (5)	98 (4)	139 (6)

- One-fourth of PLHIV initiating ART were not retained after 18 months.
- Attrition was due to treatment interruption (11%), death (10%), and migration/transfer (5%).
- Retention was higher among women compared to men (77% vs. 71%, p<0.001).

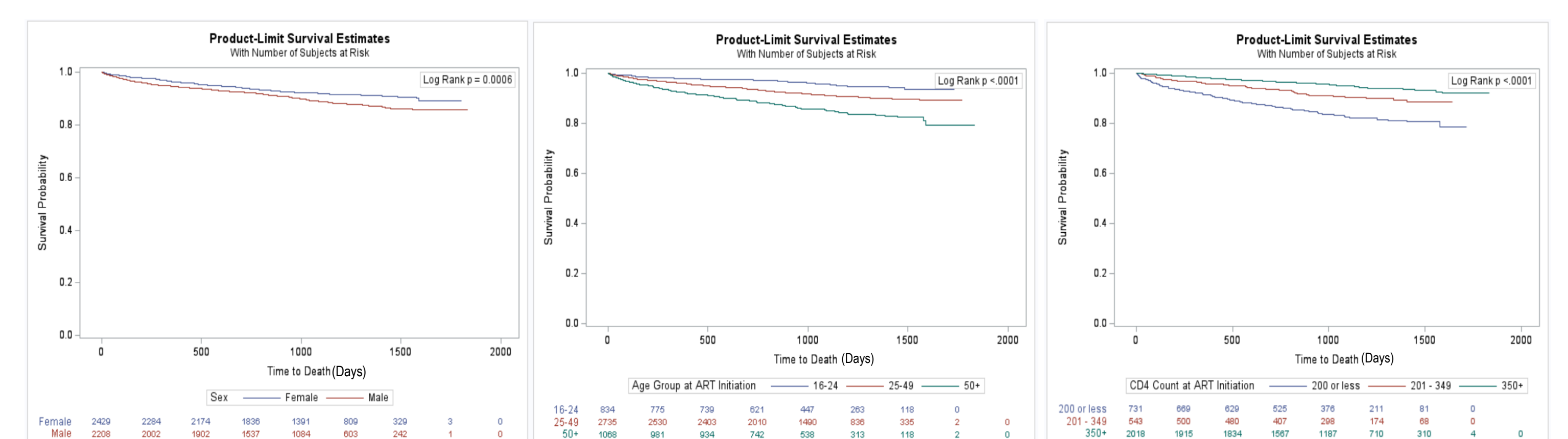
Results

Figure 2. Multivariate logistic regression: Treatment interruption amongst persons initiating ART between January 1, 2018 and December 31, 2020, Jamaica



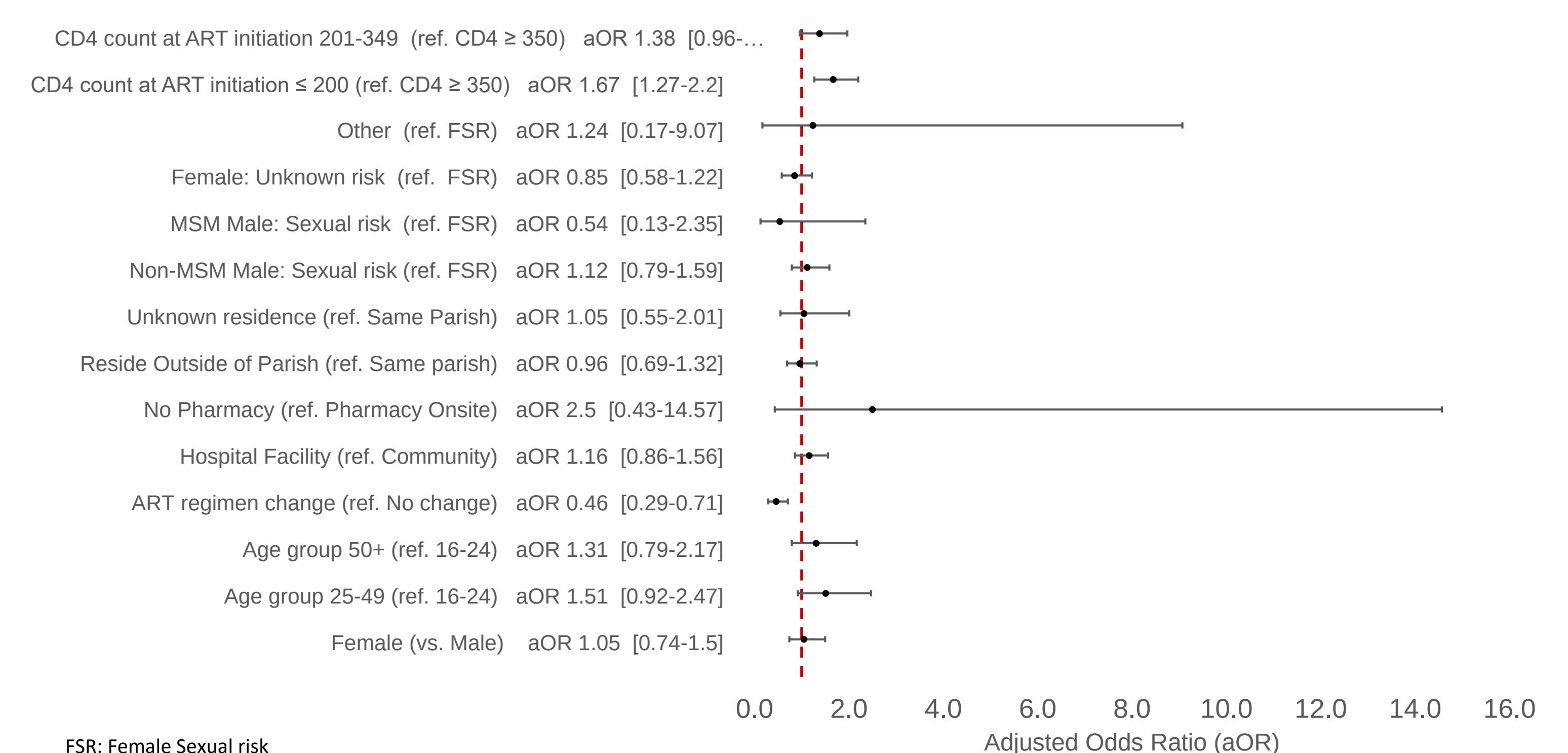
- Treatment interruption was less likely for clients aged ≥50 years, rural residents vs. urban, and clients with first CD4 count <200 cells/mm³ or 201-349 vs. those with first CD4 count ≥350 (Figure 2)

Figure 3. Kaplan-Meier survival curves for mortality by sex, age group, and CD4 count at initiation amongst persons initiating ART between January 1, 2018 and December 31, 2020, Jamaica



- Mortality rates were higher for males than females (p<0.001), persons aged 50 years and above (p<0.001) and for clients with a first CD4 count ≤200 at initiation (p<0.001) (Figure 3).

Figure 4. Adjusted results of Cox proportional hazards: Mortality amongst persons initiating ART between January 1, 2018 and December 31, 2020, Jamaica



- Treatment interruption was less likely for clients aged ≥50 years, rural residents vs. urban, and clients with first CD4 count <200 cells/mm³ or 201-349 vs. those with first CD4 count ≥350 (Figure 2)

Conclusion

Close monitoring of males, older clients, and persons initiating ART with low CD4 counts may avert mortality and reduce treatment interruptions, as well as timely review and switch for persons failing NNRTIs or other ineffective regimens

Contact Information

Corresponding Author – Dr. Sharlene Beckford Jarrett - sharlenebeck4@gmail.com